

Compre question PAPER

II SEM 2020-2021, 18 MAY 2021, FN

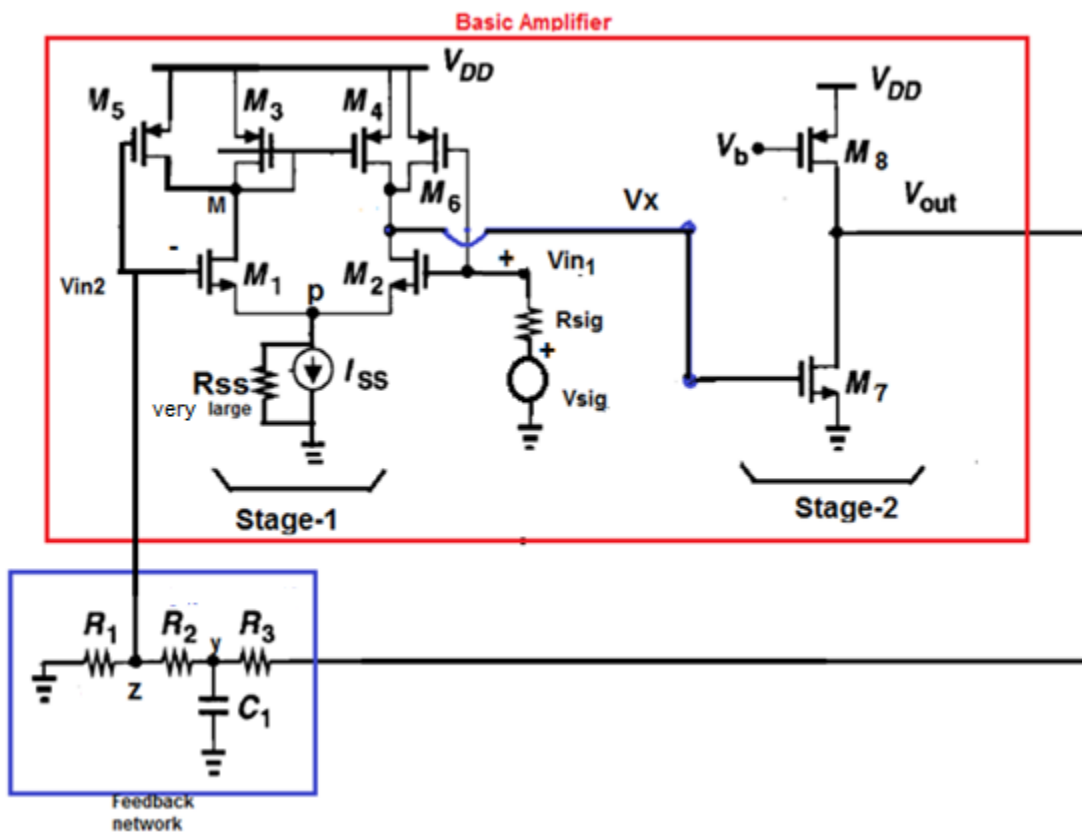
(Note: Consider values upto 2 decimal places only i.e. for example take 15.27 for value 15.277)

Q1, SET-A

Q1. Consider the circuit of feedback amplifier shown in Figure. Node z of feedback network is connected to node Vin2 of stage-1

Given:

- All transistors are properly biased in active region.
- Node p at ac ground
- for all transistor, $V_{ov} = (V_{gs} - V_t) = 0.2V$, $r_o = 1M\ \Omega$, intrinsic gain = 100
- $R_1 = 1K\Omega$, $R_2 = 9K\Omega$, $R_3 = 100K\Omega$



- a) Considering only stage-1 (without feedback network connected), determine the magnitude value of low frequency small signal voltage gain $A_1 (= \frac{V_x}{V_{in1} - V_{in2}})$ (in V/V), using its G_m and R_{out} values. [2]

Now, determine the following for entire circuit of given Figure,:

- b) Identify the type of feedback at **low frequency** . [2]
- c) Identify the type of feedback at **high frequency**. [2]
- d) Determine the magnitude value of **low frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$) , in V/V, of basic amplifier only including loading effect of feedback network. [3]
- e) Determine the magnitude value of **high frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$) , (in V/V), of basic amplifier only including loading effect of feedback network. [3]
- f) Determine the magnitude value of feedback factor β at **low frequency**. (*Note: here consider β value upto 4 decimal places only*) [1]
- g) Determine the magnitude value of feedback factor β at **high frequency**. (*Note: here consider β value upto 4 decimal places only*) [1]
- h) Determine the magnitude value of **low frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- i) Determine the magnitude value of **high frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- j) Now, Re-identify the type of feedback (at **low frequency** only) if node Vz is connected to node Vin1 and node vin2 is ac grounded. [2]

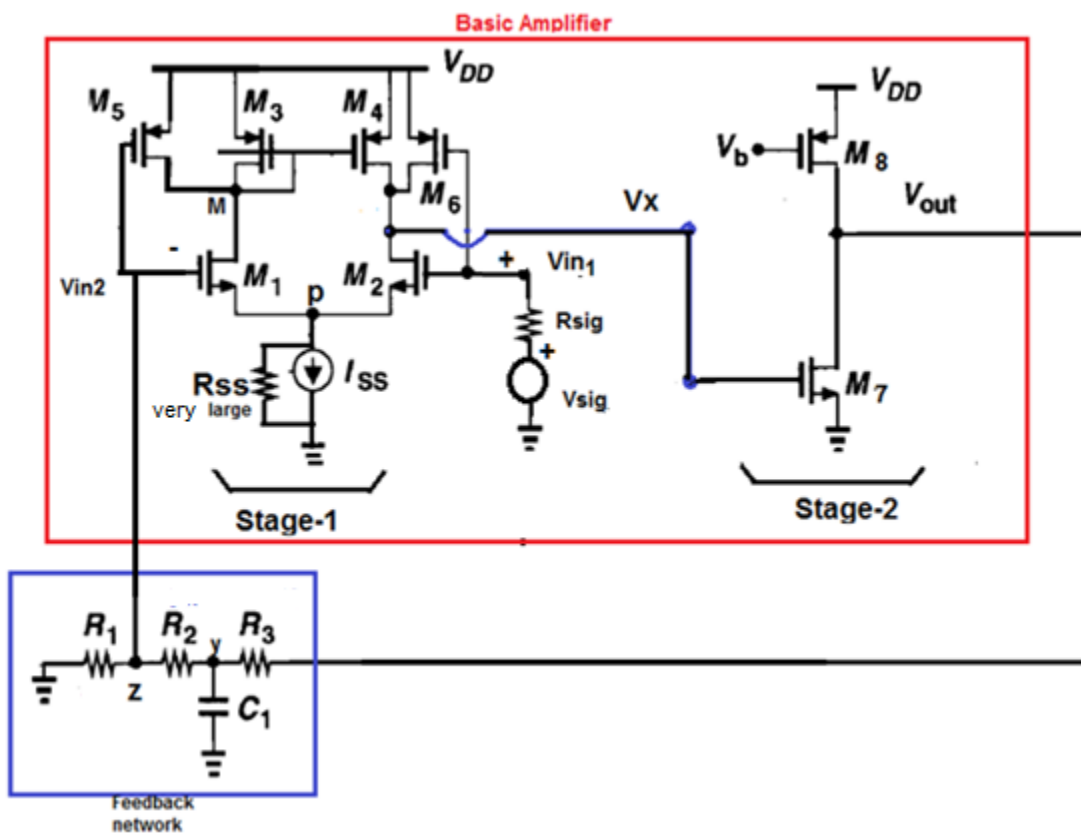
[20 marks total]

Q1, SET-B

Q1. Consider the circuit of feedback amplifier shown in Figure. Node z of feedback network is connected to node Vin2 of stage-1

Given:

- All transistors are properly biased in active region.
- Node p at ac ground
- for all transistor, $V_{ov} = (V_{gs} - V_t) = 0.2V$, $r_o = 1M\ \Omega$, intrinsic gain = 200
- $R_1 = 1K\Omega$, $R_2 = 9K\Omega$, $R_3 = 300K\Omega$



- k) Considering only stage-1 (without feedback network connected), determine the magnitude value of low frequency small signal voltage gain $A_1 (= \frac{V_x}{V_{in1} - V_{in2}})$ (in V/V), using its G_m and R_{out} values. [2]

Now, determine the following for entire circuit of given Figure,:

- l) Identify the type of feedback at **low frequency**. [2]

- m) Identify the type of feedback at **high frequency**. [2]
- n) Determine the magnitude value of **low frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$), in V/V, of basic amplifier only including loading effect of feedback network. [3]
- o) Determine the magnitude value of **high frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$), in V/V, of basic amplifier only including loading effect of feedback network. [3]
- p) Determine the magnitude value of feedback factor β at **low frequency**. (Note: here consider β value upto 4 decimal places only) [1]
- q) Determine the magnitude value of feedback factor β at **high frequency**. (Note: here consider β value upto 4 decimal places only) [1]
- r) Determine the magnitude value of **low frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- s) Determine the magnitude value of **high frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- t) Now, Re-identify the type of feedback (at **low frequency** only) if node Vz is connected to node Vin1 and node vin2 is ac grounded. [2]

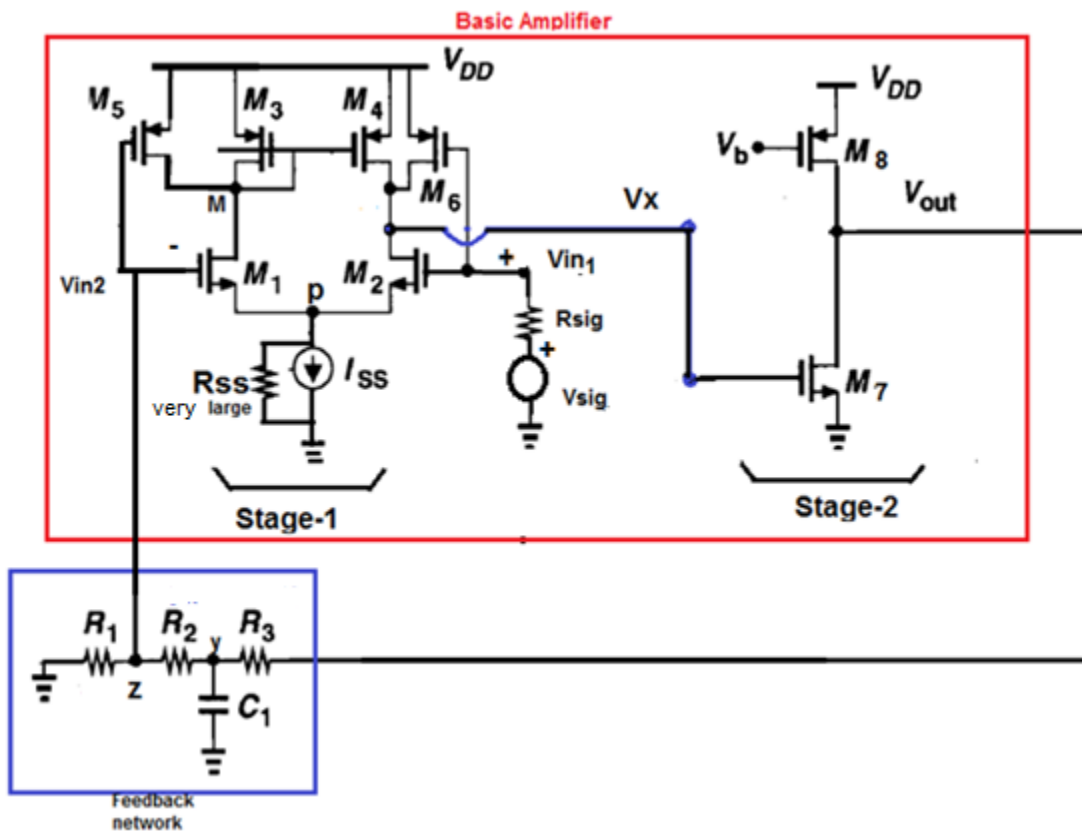
[20 marks total]

Q1, SET-C

Q1. Consider the circuit of feedback amplifier shown in Figure. Node z of feedback network is connected to node V_{in2} of stage-1

Given:

- All transistors are properly biased in active region.
- Node p at ac ground
- for all transistor, $V_{ov} = (V_{gs} - V_t) = 0.2V$, $r_o = 1M\ \Omega$, intrinsic gain = 400
- $R_1 = 1\ K\Omega$, $R_2 = 9\ K\Omega$, $R_3 = 500\ K\Omega$



- u) Considering only stage-1 (without feedback network connected), determine the magnitude value of low frequency small signal voltage gain $A_1 (= \frac{V_x}{V_{in1} - V_{in2}})$ (in V/V), using its G_m and R_{out} values. [2]

Now, determine the following for entire circuit of given Figure,:

- v) Identify the type of feedback at **low frequency**. [2]
- w) Identify the type of feedback at **high frequency**. [2]

- x) Determine the magnitude value of **low frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$), in V/V, of basic amplifier only including loading effect of feedback network. [3]
- y) Determine the magnitude value of **high frequency** small signal gain ($A = \frac{V_{out}}{V_{in1} - V_{in2}}$), in V/V, of basic amplifier only including loading effect of feedback network. [3]
- z) Determine the magnitude value of feedback factor β at **low frequency**. (Note: here consider β value upto 4 decimal places only) [1]
- aa) Determine the magnitude value of feedback factor β at **high frequency**. (Note: here consider β value upto 4 decimal places only) [1]
- bb) Determine the magnitude value of **low frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- cc) Determine the magnitude value of **high frequency** small signal closed loop gain $A_f = \left[\frac{V_{out}}{V_{sig}} \right]$ (in V/V),. [2]
- dd) Now, Re-identify the type of feedback (at **low frequency** only) if node Vz is connected to node Vin1 and node vin2 is ac grounded. [2]

[20 marks total]

Q2

(Note: Consider values upto 2 decimal places only i.e. for example take 15.28 for value 15.277)

Q2. Consider the circuit shown in Fig. , $V_{out} = V_{out1} - V_{out2}$. Assume matching of transistor wherever necessary. Assume no mismatch in the circuit

Assume proper active region DC bias of all transistors with $r_o = 1M\Omega$, $V_{ov} = (V_{gs} - V_t) = 0.2V$.

Neglect other parasitic capacitances. Iss is basic current mirror circuit.

a) If the very low frequency small signal voltage gain $[V_{out}/V_{in}] = 10^6$, determine the values of intrinsic gain $(g_m r_o)$, and g_m .

b) Determine the value of small signal impedances at node Y at low frequency.

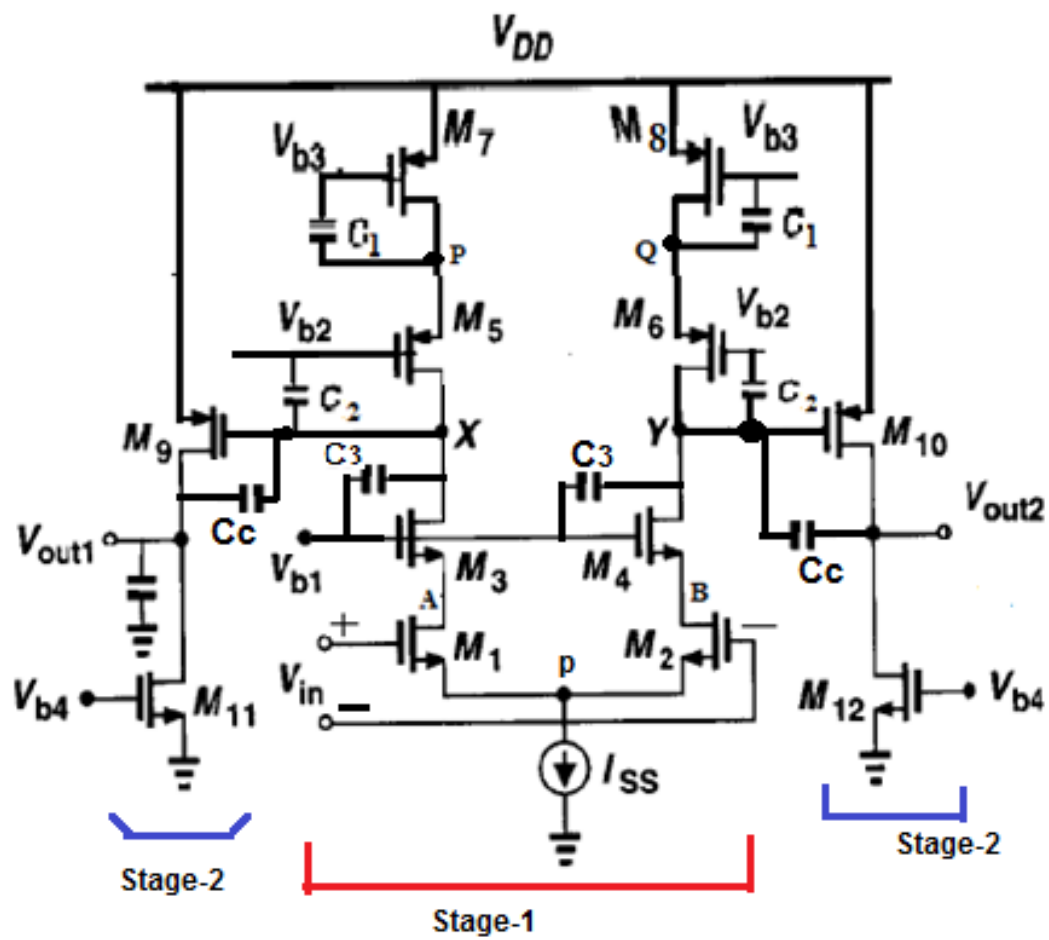
c) Determine the value of small signal voltage gain $[V_{out}/V_{in}]$.

- i. if only C_c starts conducting.
- ii. if only C_3 starts conducting.
- iii. if only C_1 starts conducting.
- iv. if only C_2 starts conducting.

d) Given $C_c = 10C_1$, $C_1 = 10C_2$, $C_2 = C_3$, and capacitor C_c starts conducting at $\omega \geq 10^4$ rad/sec.

Calculate the value of magnitude of small signal voltage gain $[V_{out}/V_{in}]$ (in dB) at frequencies-- ω_{CC} , ω_{c1} , ω_{c2} , ω_{c3} ?

(Note: write asymptotic values of magnitude of small signal voltage gain only)



[20M)

Q3

SET A

(Note: Consider values upto 2 decimal places only in calculation i.e. for example take 15.27 for value 15.277)

Unless given specifically

Take -- $V_{DD} = 4V$,

For NMOS device $\mu_n C_{ox} = 140 \mu A/V^2$, $V_T = 0.8 V$, $\lambda = 0.1 V^{-1}$, $\gamma = 0 \sqrt{V}$

For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.8 V$, $\lambda = 0.1 V^{-1}$, $\gamma = 0 \sqrt{V}$

For NPN/ PNP device $\beta = 100$, $V_{CESAT} = 0.2V$, $V_A = 100V$, $KT/q = 25mV$ (at room temp.), $I_s = 10^{-14} A$

NOTE:

If not specified in question,-----

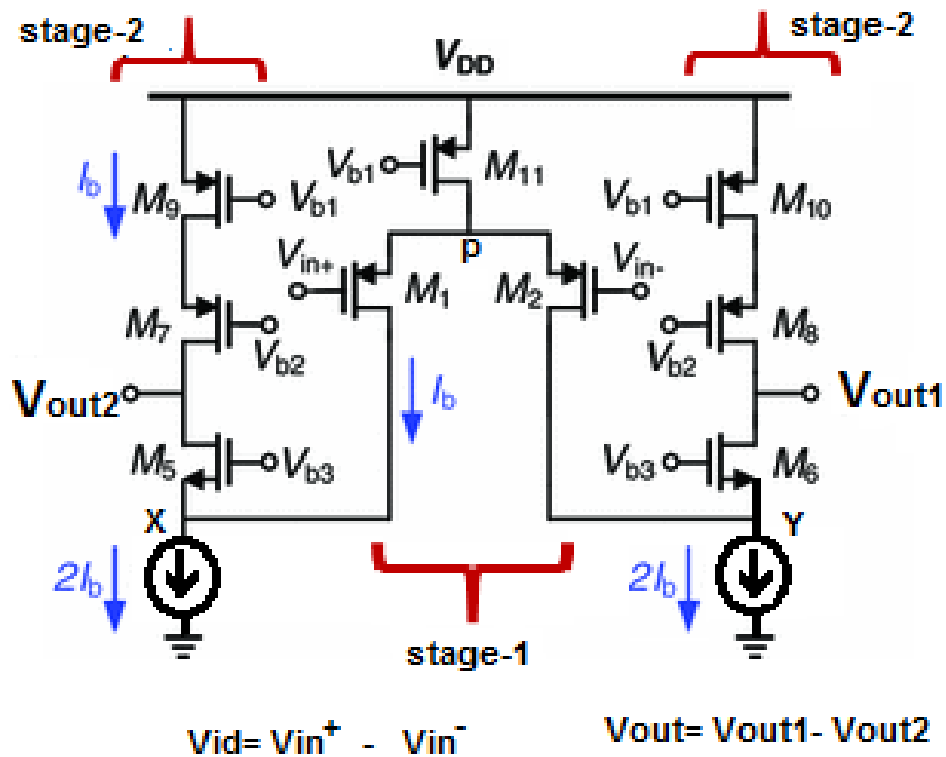
- Ignore γ , λ in drain current equation. Assume matched components wherever required
- Bulk of NMOS connected to ground and bulk of PMOS connected to Vdd. Specify your assumptions. Justify your answers.
- Unless specified in the question, assume all MOSFET are biased in saturation region.
- All symbols have usual meaning. Assume matching of transistors if necessary

Note: Cheating cases shall be dealt with strictly.

Q1. Consider the circuit of operational amplifier shown in **Figure**.

Given:

- There is no mismatch in transistor pairs wherever matching is required.
- All transistors are properly biased in active region.
- Node p at ac ground. Current source of value $(2I_b)$ is a basic current mirror circuit.
- For all transistor, $V_{dd} = 4V$, $V_{ov} = (V_{gs} - V_t) = 0.2V$, $r_o = 1M \Omega$, intrinsic gain = 100



- Identify amplifiers of stage-1, and stage-2. [4]
- Determine the value of DC current I_b (in μA). [2]
- Determine output voltage swing for differential output 'Vout' in (V) [4]
- Determine the value of low frequency differential mode small signal gain $A_{dm} = \frac{V_{out}}{V_{id}}$ (in V/V) [3]
- Determine the value of low frequency common mode rejection ratio at low frequency. There is no mismatch in transistor pairs. [2]

---- 15 marks total

Q3, SET-B

(Note: Consider values upto 2 decimal places only in calculation i.e. for example take 15.27 for value 15.277)

Unless given specifically

Take -- $V_{DD} = 4V$,

For NMOS device $\mu_n C_{ox} = 140 \mu A/V^2$, $V_T = 0.8 V$, $\lambda = 0.1 V^{-1}$, $\gamma = 0 \sqrt{V}$

For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.8 V$, $\lambda = 0.1 V^{-1}$, $\gamma = 0 \sqrt{V}$

For NPN/ PNP device $\beta = 100$, $V_{CESAT} = 0.2V$, $V_A = 100V$, $KT/q = 25mV$ (at room temp.), $I_s = 10^{-14} A$

NOTE:

If not specified in question,-----

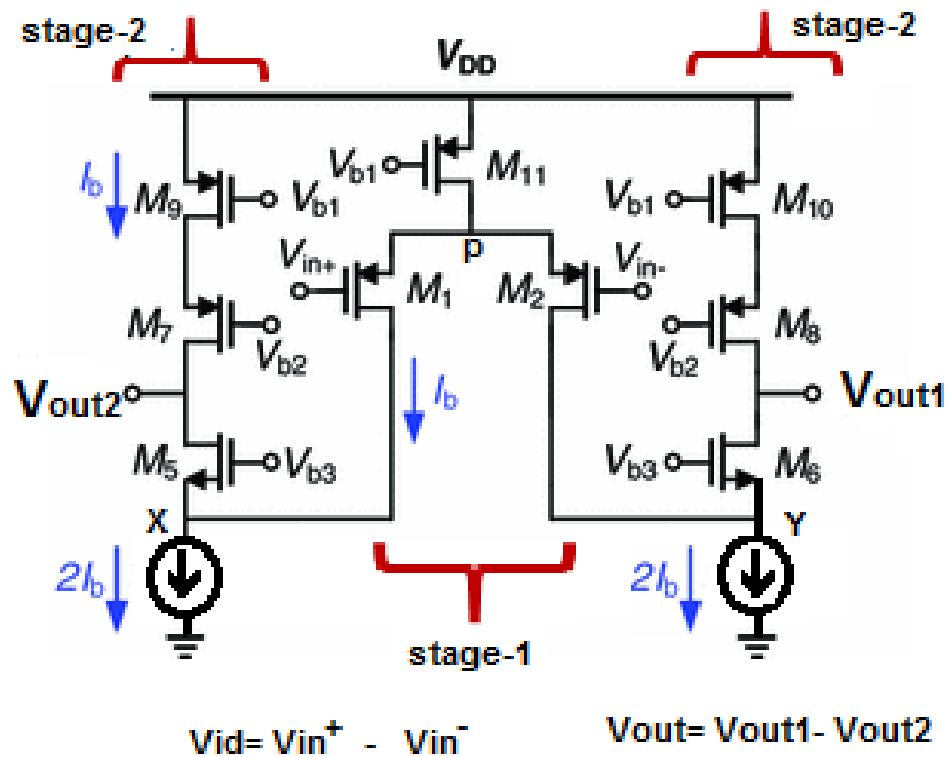
- Ignore γ , λ in drain current equation. Assume matched components wherever required
- Bulk of NMOS connected to ground and bulk of PMOS connected to Vdd. Specify your assumptions. Justify your answers.
- Unless specified in the question, assume all MOSFET are biased in saturation region.
- All symbols have usual meaning. Assume matching of transistors if necessary

Note: Cheating cases shall be dealt with strictly.

Q1. Consider the circuit of operational amplifier shown in Figure.

Given:

- There is no mismatch in transistor pairs wherever matching is required.
- All transistors are properly biased in active region.
- Node p at ac ground. Current source of value ($2I_b$) is a basic current mirror circuit.
- For all transistor, $V_{dd} = 4.5V$, $V_{ov} = (V_{gs} - V_t) = 0.3V$, $r_o = 1M \Omega$, intrinsic gain = 200



- f) Identify amplifiers of stage-1, and stage-2. [4]
- g) Determine the value of DC current I_b (in μA). [2]
- h) Determine output voltage swing for differential output 'Vout' in (V) [4]
- i) Determine the value of low frequency differential mode small signal gain $A_{dm} = \frac{V_{out}}{V_{id}}$ (in V/V) [3]
- j) Determine the value of low frequency common mode rejection ratio at low frequency. [2]

---- 15 marks total

Q4 SET A

Q.4 Consider the circuit shown in Fig. 1. Given $V_{DD} = 10\text{ V}$, transistor parameters $\beta = \infty$, $V_A = \infty$, and $V_{BE(on)} = V_{EB(on)} = 0.7\text{ V}$. Let $R_4 = 24\text{ k}\Omega$.

(a) Find I_{REF} , I_1 , I_2 , and I_3 .

(b) Determine the maximum values of R_1 , R_2 , and R_3 such that Q_1 , Q_2 , and Q_3 remain biased in the forward-active region.

Assume $V_{CE(min)} = V_{EC(min)} = 0.7\text{ V}$.

(2+1+1+1+1+1+1 = 8 marks)

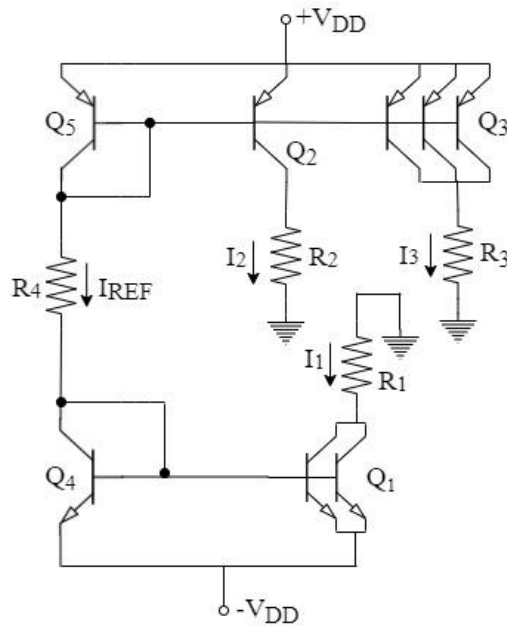


Fig. 1

Q4, SET B

Q.4 Consider the circuit shown in Fig. 1. Given $V_{DD} = 5\text{ V}$, transistor parameters $\beta = \infty$, $V_A = \infty$, and $V_{BE(on)} = V_{EB(on)} = 0.7\text{ V}$. Let $R_4 = 18\text{ k}\Omega$. (a) Find I_{REF} , I_1 , I_2 , and I_3 . (b) Determine the maximum values of R_1 , R_2 , and R_3 such that Q_1 , Q_2 , and Q_3 remain biased in the forward-active region. Assume $V_{CE(min)} = V_{EC(min)} = 0.7\text{ V}$. (2+1+1+1+1+1+1 = 8 marks)

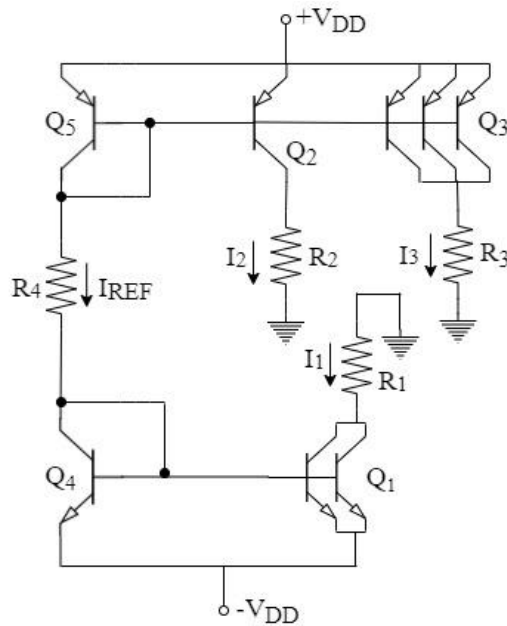


Fig. 1

Q4 – SET C

Q.4 Consider the circuit shown in Fig. 1. Given $V_{DD} = 7\text{ V}$, transistor parameters $\beta = \infty$, $V_A = \infty$, and $V_{BE(on)} = V_{EB(on)} = 0.7\text{ V}$. Let $R_4 = 12\text{ k}\Omega$. (a) Find I_{REF} , I_1 , I_2 , and I_3 . (b) Determine the maximum values of R_1 , R_2 , and R_3 such that Q_1 , Q_2 , and Q_3 remain biased in the forward-active region. Assume $V_{CE(min)} = V_{EC(min)} = 0.7\text{ V}$. (2+1+1+1+1+1+1 = 8 marks)

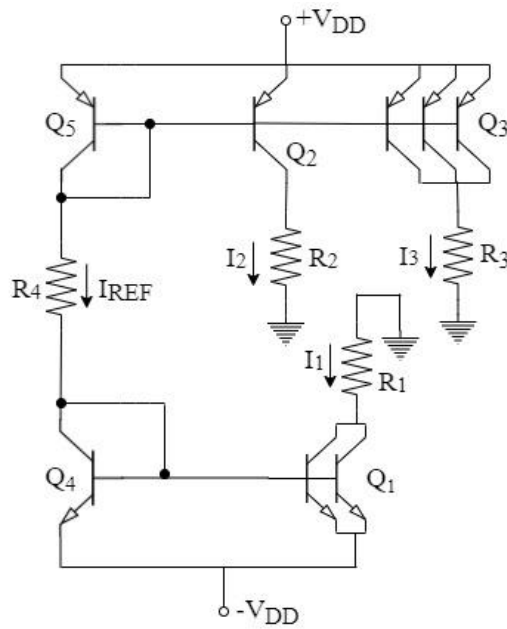


Fig. 1

Q5, SET A

Q.5 Consider a cascode amplifier with cascode active load as shown in Fig. 2.

Given $V_{DD} = 10 \text{ V}$, $I_{REF} = 200 \mu\text{A}$, transistor parameters are given as

$V_{TN} = 0.8 \text{ V}$, $V_{TP} = -0.8 \text{ V}$, $\mu_n C_{ox} = 140 \mu\text{A/V}^2$, $\mu_p C_{ox} = 40 \mu\text{A/V}^2$, and $\lambda_n = \lambda_p = 0.01 \text{ V}^{-1}$.

The **W/L** ratios of **M1 and M2** are **10**, and those of all **other transistors** are **20**.

The value of V_{GSQ} is such that drain current of M1, $I_{D1} = 200 \mu\text{A}$, and all transistors are biased in the saturation region.

Determine small signal parameters g_{m1} , g_{m3} , r_{o1} , r_{o3} , R_{EFF} (effective resistance looking into the drain of M3), small signal voltage gain $A_v = v_o/v_i$. (1+1+1+1+3+3 = 10 marks)

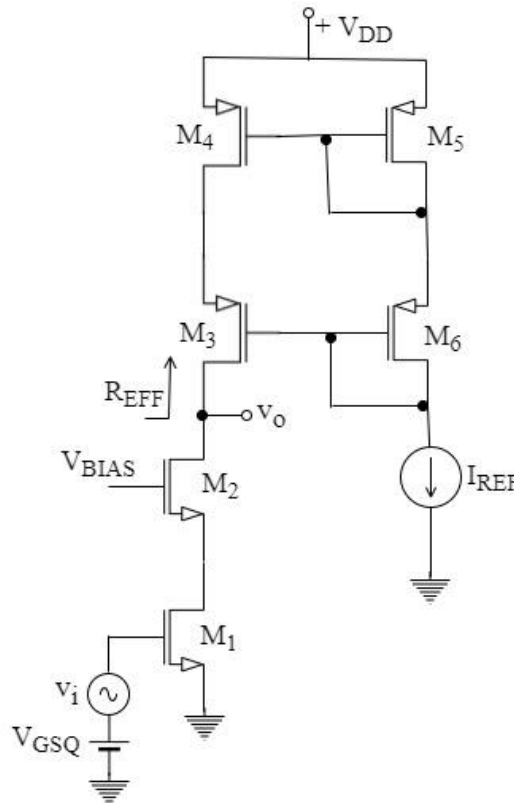


Fig. 2

Q5, SET-B

Q.5 Consider a cascode amplifier with cascode active load as shown in Fig. 2. Given $V_{DD} = 8\text{ V}$, $I_{REF} = 400\text{ }\mu\text{A}$ transistor parameters are given as $V_{TN} = 0.8\text{ V}$, $V_{TP} = -0.8\text{ V}$, $\mu_n C_{ox} = 140\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox} = 40\text{ }\mu\text{A/V}^2$, and $\lambda_n = \lambda_p = 0.01\text{ V}^{-1}$. The W/L ratios of **M1** and **M2** are **20**, and those of all **other transistors** are **40**. The value of V_{GSQ} is such that drain current of **M1**, $I_{D1} = 400\text{ }\mu\text{A}$, and all transistors are biased in the saturation region. Determine small signal parameters g_{m1} , g_{m3} , r_{o1} , r_{o3} , R_{EFF} (effective resistance looking into the drain of **M3**), small signal voltage gain $A_v = v_o/v_i$. (1+1+1+1+3+3 = 10 marks)

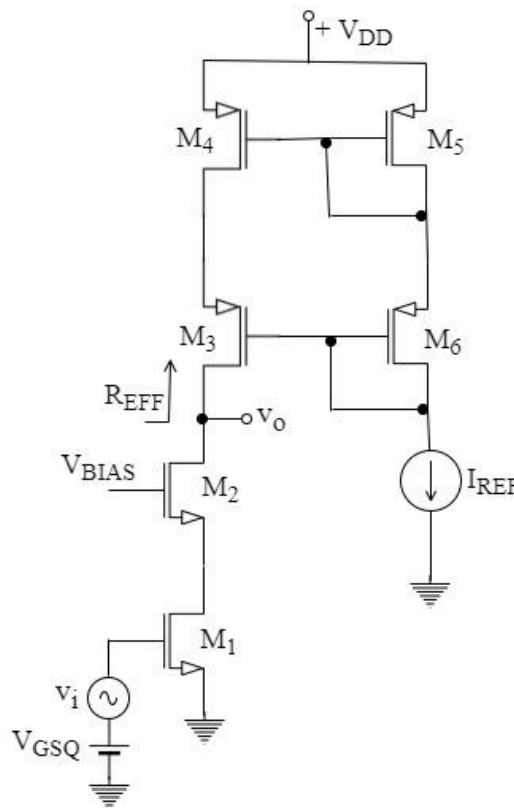


Fig. 2

Q5, SET-C

Q.2 Consider a cascode amplifier with cascode active load as shown in Fig. 2. Given $V_{DD} = 12\text{ V}$, $I_{REF} = 250\text{ }\mu\text{A}$, transistor parameters are given as $V_{TN} = 0.8\text{ V}$, $V_{TP} = -0.8\text{ V}$, $\mu_n C_{ox} = 140\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox} = 40\text{ }\mu\text{A/V}^2$, and $\lambda_n = \lambda_p = 0.01\text{ V}^{-1}$. The W/L ratios of **M1** and **M2** are **12**, and those of all **other transistors** are **24**. The value of V_{GSQ} is such that drain current of **M1**, $I_{D1} = 250\text{ }\mu\text{A}$, and all transistors are biased in the saturation region. Determine small signal parameters g_{m1} , g_{m3} , r_{o1} , r_{o3} , R_{EFF} (effective resistance looking into the drain of **M3**), small signal voltage gain $A_v = v_o/v_i$. (1+1+1+1+3+3 = 10 marks)

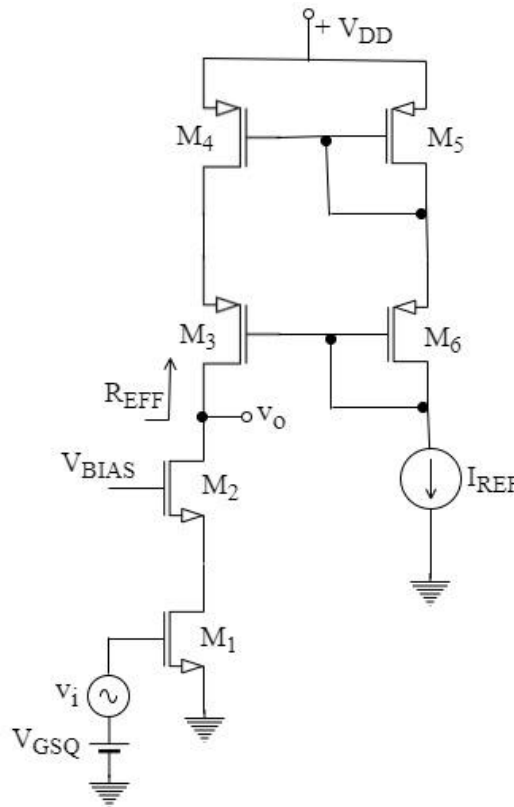
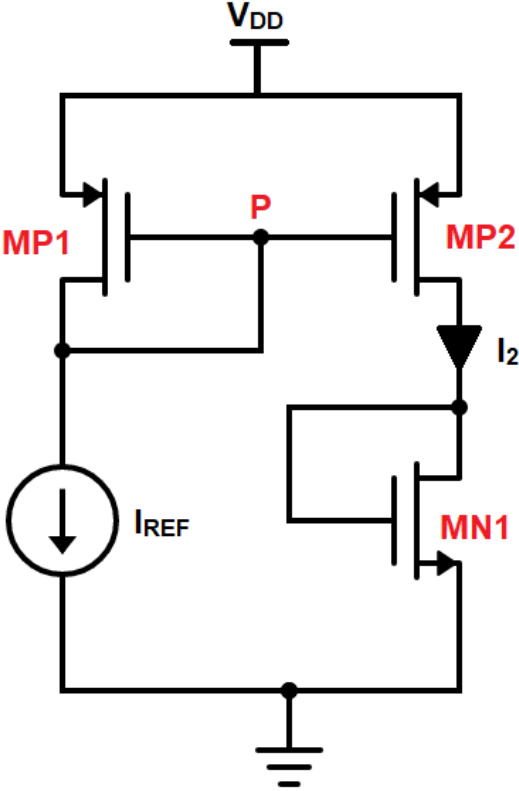


Fig. 2

Q6—Q9

Q6 [3 Marks] A current mirror circuit is shown in figure. Given the following data, $V_{DD} = 3V$, $I_{REF} = 0.2 \text{ mA}$, $V_{T0n} = |V_{T0p}| = 0.8V$, $\mu_n C_{ox} = 3 \mu_p C_{ox} = 0.6 \text{ mA/V}^2$. Ignore channel length modulation effect. Compute the drain current of the NMOS (I_2). (Answer in mA).

Set-A: Aspect ratio (W/L) of MP1, MP2 and MN1 are 5,15 and 10 respectively. Answer: 0.6 mA Set-B: Aspect ratio (W/L) of MP1, MP2 and MN1 are 2,10 and 10 respectively. Answer: 1.0 mA Set-C: Aspect ratio (W/L) of MP1, MP2 and MN1 are 3,12 and 20 respectively. Answer: 0.8 mA	
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Q7 [2+2 Marks] A common source amplifier, can be represented as a single pole low pass response circuit. The maximum voltage gain is **30/40/45 dB** and the pole frequency is **1/5/10 Krad/s**. Given the output waveform is $V_p \cos(\omega t - 55^\circ)$ when the input is $20 \cos(\omega t) \text{ mV}$. Find the Value of V_p (in mV) and ω (in Krad/sec).

Solution:

Transfer function of a LPF = $K/(s+p)$

Max. Voltage gain = 30dB = 31.627 V/V.

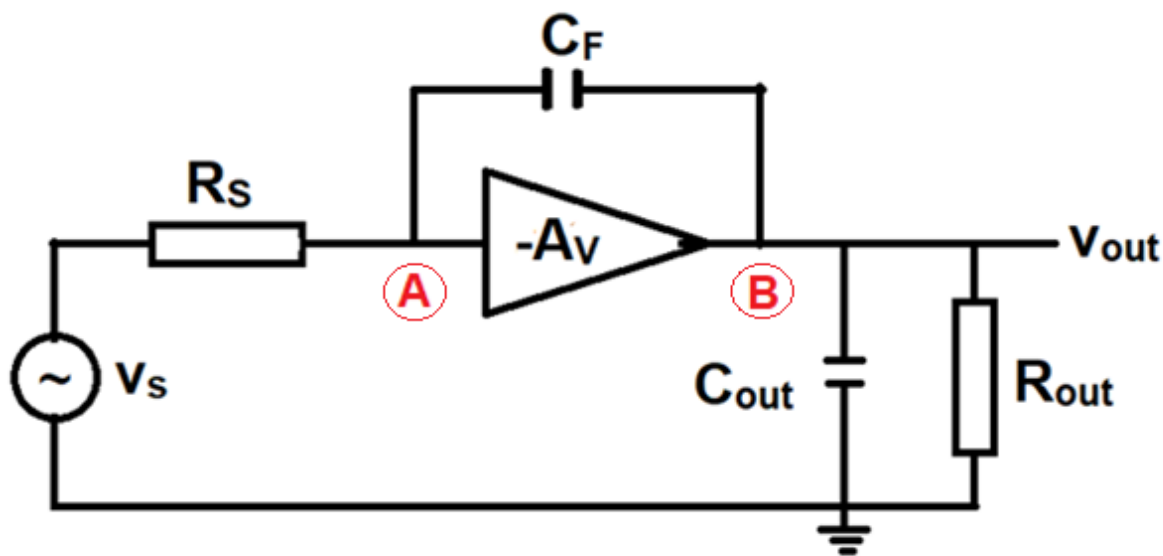
Phase shift = $-\tan^{-1}(\omega/\omega_p) = -55^\circ$ $\tan(55) = 1.4281$.

Set-A: (30dB, 1Krad/s) $\omega = 1.4281$ krad/sec $V_P = 362.81$ mV

Set-B: (40dB, 5Krad/s) $\omega = 7.1405$ krad/sec $V_P = 1147.15$ mV

Set-C: (50dB, 10Krad/s) $\omega = 14.281$ krad/sec $V_P = 2039.958$ mV

Q8 [2+2 Marks] For the circuit shown in figure below. The amplifier has a gain of $A_v = 30/40/50$. The negative sign before A_v indicates the phase shift. The Amplifier has a very high input impedance ($\sim$ infinite) and a low output impedance (~ 0). Given $R_{out} = 10\text{k}\Omega$, $R_S = 20\text{k}\Omega$, $C_{out} = 0.2\text{pF}$, $C_F = 50/60/70\text{fF}$, using Millers Theorem, estimate the magnitude of the poles at node-A and node-B in Mrad/s/



Solution: Resistance at node-A R_S and Resistance at node-B R_{out}

Miller Cap. At node-A is $C_F(1+A_v)$

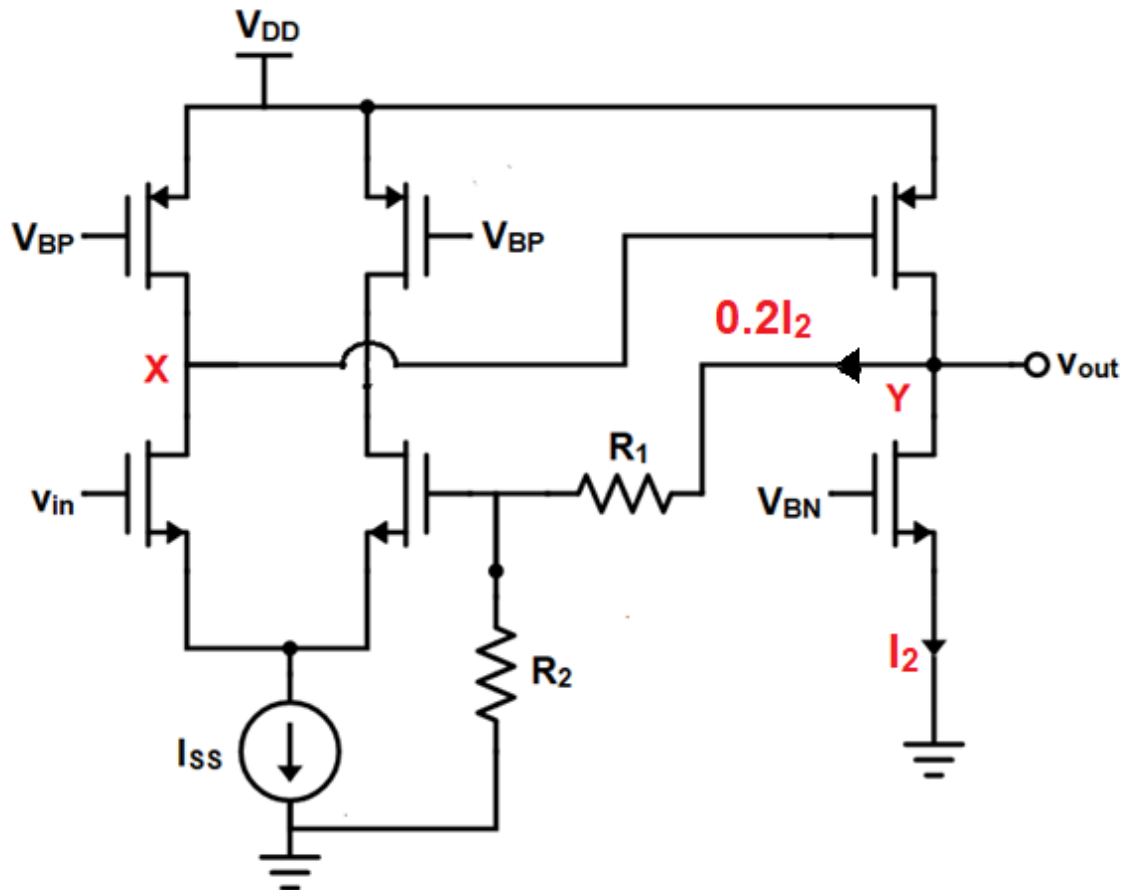
Miller Cap. At node-B is $C_F(1+(1/A_v))$

Set-A: (30,50f) Pole-A = 64.516 Mrad/s Pole-B = 397.350 Mrad/sec

Set-B: (40,60f) Pole-A = 26.881 Mrad/s Pole-B = 382.409 Mrad/sec

Set-C: (50,70f) Pole-A = 14.005 Mrad/s Pole-B = 368.459 Mrad/sec

Q9 [3+3 Marks] A MOSFET based Amplifier is shown in fig. All MOSFETS are in saturation. Neglect body bias effect, For all the MOSFETS $\lambda = 0.01/\text{V}$,
 Equivalent Cap. At node X & Y: $C_X = 500\text{fF}$, $C_Y = 1\text{pF}$.
 $R_1 = 40\text{k}\Omega$, $R_2 = 10\text{k}\Omega$.
 $I_{SS} = I_2 = 0.2/0.3/0.4 \text{ mA}$
 Compute the poles (magnitude only) at node-X and node-Y in Mrad/s.



Solution: $r_0 = 1/(\lambda I_D)$

Resistance at Node-X = $r_0/2$ (Same current through NMOS and PMOS $I_{SS}/2$)

Resistance at Node-Y = $(R_1 + R_2) \parallel (r_{0n}) \parallel (r_{0p})$

In second stage current differs for NMOS and PMOS

NMOS current = I_2 PMOS current = $1.2 \cdot I_2$

Set-A: (0.2mA) Pole-X = 4 Mrad/s Pole-Y = 24.4 Mrad/sec

Set-B: (0.3mA) Pole-X = 6 Mrad/s Pole-Y = 26.6 Mrad/sec

Set-C: (0.4mA) Pole-X = 8 Mrad/s Pole-Y = 28.8 Mrad/sec

Q10----SET A

Consider the circuit in the below figure.

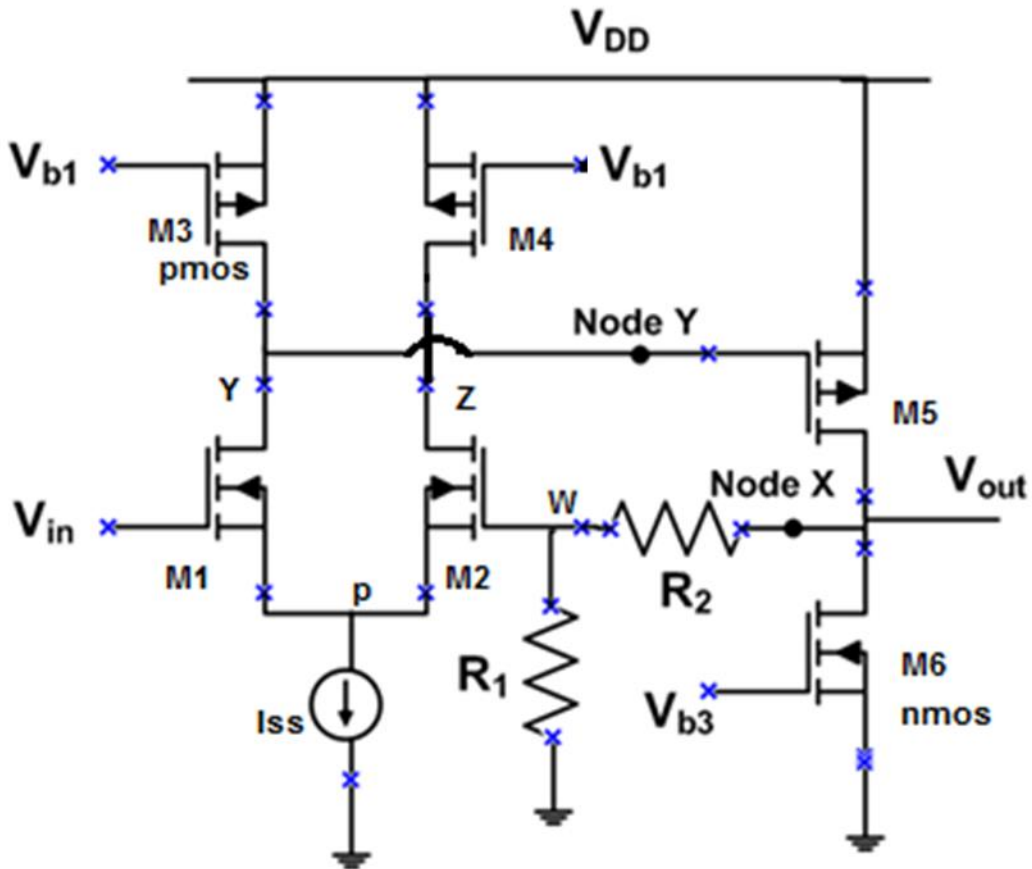
Assume all transistors are properly biased and operating in active region. I_{ss} is current source with high impedance. Node P at ac ground.

Assume all transistors are properly biased and operating in active region. I_{ss} is current source with high impedance. Node P at ac ground.

Given: $\lambda = 0.01V^{-1}$, $I_{ss} = 1mA$, $g_m = 2mA/V$.

Low frequency gain= $A_M = \frac{V_{out}}{V_{in}} = 92dB$ and the total capacitance at node X and Y is $0.5pF$.

Take $\omega_{px} = \omega_{py} = -20 M rad/sec$.



Determine the following:

- What is the unity gain bandwidth 'UGB'(rad/sec)? (2 Marks)
- Phase margin for unity gain feedback (in degree). (2 Marks)
- Required value of Unity gain bandwidth (UGB') for phase margin of 60° (UGB').
Location of phase cross over frequency remains constant. (2 Marks)

- d) The modified value of pole frequency ω_{py}' (in radian/ sec) to obtain the value of unity gain bandwidth (UGB') in part (c). (2 Marks)
- e) Compensating capacitor C_C required between node Y and ground to obtain ω_{py}' . (2 Marks)
- f) The value of the Miller capacitance C_M required between node Y and node X to obtain ω_{py}' . (2 Marks)
- g) The minimum value of resistance R_z needed to ensure a negative zero frequency due to Miller capacitance C_M (2 Marks)
-

Q10----SET B

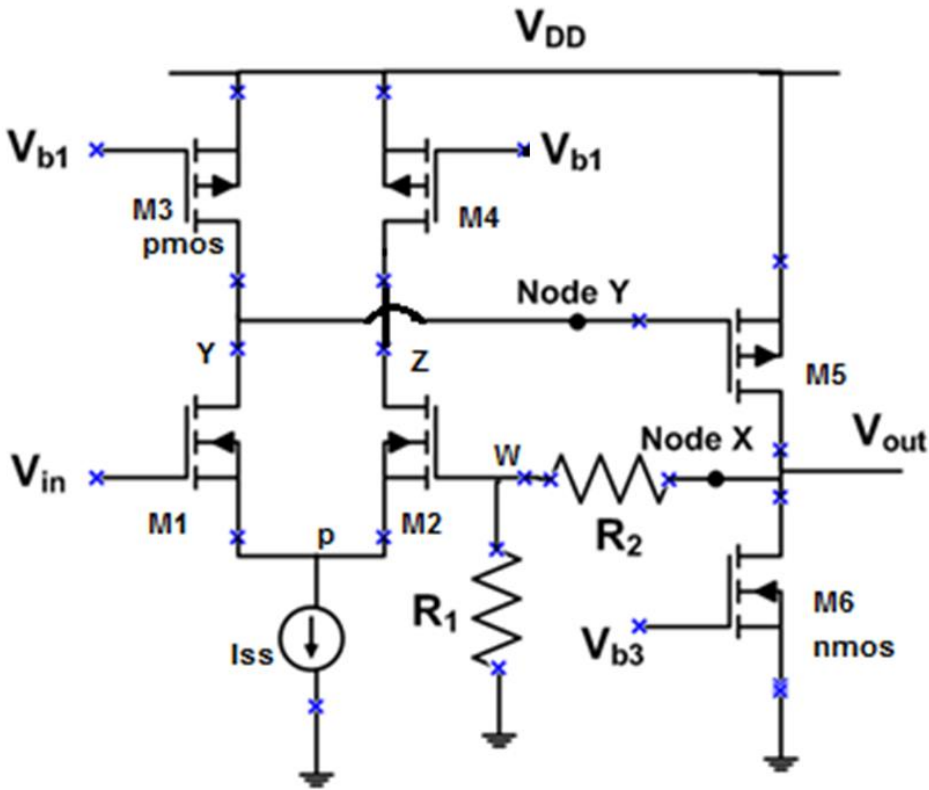
Consider the circuit in the below figure.

Assume all transistors are properly biased and operating in active region. I_{ss} is current source with high impedance. Node P at ac ground.

Given: $\lambda = 0.01V^{-1}$, $I_{SS} = 1mA$, $g_m = 3mA/V$.

Low frequency gain= $A_M = \frac{V_{out}}{V_{in}} = 92dB$ and the total capacitance at node X and Y is 0.3pF.

Take $\omega_{px} = \omega_{py} = -50 M rad/sec$.



Determine the following:

- What is the unity gain bandwidth 'UGB'(rad/sec)? **(2 Marks)**
- Phase margin for unity gain feedback (in degree). **(2 Marks)**
- Required value of Unity gain bandwidth (**UGB'**) for phase margin of 70° (**UGB'**).
Location of phase cross over frequency remains constant. **(2 Marks)**
- The modified value of pole frequency ω_{py}' (**in radian/ sec**) to obtain the value of unity gain bandwidth (**UGB'**) in part (c). **(2 Marks)**
- Compensating capacitor C_C required between node Y and ground to obtain ω_{py}' . **(2 Marks)**
- The value of the Miller capacitance C_M required between node Y and node X to obtain ω_{py}' . **(2 Marks)**
- The minimum value of resistance R_Z needed to ensure a negative zero frequency due to Miller capacitance C_M **(2 Marks)**

- What is the unity gain bandwidth 'UGB' (rad/sec)? **(2 Marks)**
- Phase margin for unity gain feedback (in degree). **(2 Marks)**
- Required value of Unity gain bandwidth (**UGB'**) for phase margin of 70° (**UGB'**).
Location of phase cross over frequency remains constant. **(2 Marks)**
- The modified value of pole frequency **ω_{py} ' (in radian/ sec)** to obtain the value of unity gain bandwidth (**UGB'**) in part (c). **(2 Marks)**

- e) Compensating capacitor C_C required between node Y and ground to obtain ω_{py}' . (2 Marks)
- f) The value of the Miller capacitance C_M required between node Y and node X to obtain ω_{py}' . (2 Marks)
- g) The minimum value of resistance R_z needed to ensure a negative zero frequency due to Miller capacitance C_M (2 Marks)

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